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1  -- =====
2  -- Part 01 - Online Retail Store Database
3  -- Builds the database from the provided relational schema, then adds a few
4  -- sample rows so the Part 02 operations have data to work on.
5  -- Dialect: T-SQL (SQL Server)
6  -- =====
7
8  USE master;
9  GO
10
11 IF DB_ID('OnlineRetailStore') IS NOT NULL
12 BEGIN
13     ALTER DATABASE OnlineRetailStore SET SINGLE_USER WITH ROLLBACK IMMEDIATE;
14     DROP DATABASE OnlineRetailStore;
15 END
16 GO
17
18 CREATE DATABASE OnlineRetailStore;
19 GO
20
21 USE OnlineRetailStore;
22 GO
23
24 -- MainCategory is self-referencing, for subcategories (e.g. "Laptops" under "Electronics").
25 CREATE TABLE Categories (
26     CategoryId      INT IDENTITY(1,1) PRIMARY KEY,
27     Name            NVARCHAR(100) NOT NULL,
28     Description      NVARCHAR(500) NULL,
29     MainCategory     INT NULL,
30     CONSTRAINT FK_Categories_MainCategory FOREIGN KEY (MainCategory)
31         REFERENCES Categories (CategoryId)
32 );
33
34 CREATE TABLE Suppliers (
35     SupplierId      INT IDENTITY(1,1) PRIMARY KEY,
36     Name            NVARCHAR(150) NOT NULL,
37     Country         NVARCHAR(100) NULL,
38     Email           NVARCHAR(150) NULL,
39     Address         NVARCHAR(250) NULL,
40     ContactNumber   NVARCHAR(20) NULL
41 );
```

```
43 -- CategoryId allows NULL so a product can be added before it is assigned a category.
44 CREATE TABLE Products (
45     ProductId      INT IDENTITY(1,1) PRIMARY KEY,
46     StockQuantity  INT NOT NULL DEFAULT 0,
47     Name           NVARCHAR(150) NOT NULL,
48     AddedDate      DATETIME2 NOT NULL DEFAULT SYSDATETIME(),
49     Description     NVARCHAR(1000) NULL,
50     UnitPrice      DECIMAL(10, 2) NOT NULL,
51     CategoryId     INT NULL,
52     CONSTRAINT FK_Products_Categories FOREIGN KEY (CategoryId)
53         REFERENCES Categories (CategoryId)
54 );
55
56 CREATE TABLE Products_Suppliers (
57     SupplierId     INT NOT NULL,
58     ProductId      INT NOT NULL,
59     CONSTRAINT PK_Products_Suppliers PRIMARY KEY (SupplierId, ProductId),
60     CONSTRAINT FK_Products_Suppliers_Suppliers FOREIGN KEY (SupplierId)
61         REFERENCES Suppliers (SupplierId),
62     CONSTRAINT FK_Products_Suppliers_Products FOREIGN KEY (ProductId)
63         REFERENCES Products (ProductId)
64 );
65
66 CREATE TABLE StockTransactions (
67     TranId         INT IDENTITY(1,1) PRIMARY KEY,
68     TranDate       DATETIME2 NOT NULL DEFAULT SYSDATETIME(),
69     QuantityChange INT NOT NULL,
70     Type           NVARCHAR(3) NOT NULL,
71     Reference      INT NULL,
72     ProductId      INT NOT NULL,
73     CONSTRAINT FK_StockTransactions_Products FOREIGN KEY (ProductId)
74         REFERENCES Products (ProductId),
75     CONSTRAINT CK_StockTransactions_Type CHECK (Type IN ('in', 'out'))
76 );
77
78 CREATE TABLE Customers (
79     CustomerId     INT IDENTITY(1,1) PRIMARY KEY,
80     FullName       NVARCHAR(150) NOT NULL,
81     PhoneNumber    NVARCHAR(20) NULL,
82     Email          NVARCHAR(150) NOT NULL UNIQUE,
83     ShippingAddress NVARCHAR(250) NULL,
84     RegistrationDate DATETIME2 NOT NULL DEFAULT SYSDATETIME()
85 );
```

```
87 CREATE TABLE Orders (  
88     OrderId      INT IDENTITY(1,1) PRIMARY KEY,  
89     Status       NVARCHAR(50) NOT NULL,  
90     TotalAmount  DECIMAL(10, 2) NOT NULL,  
91     OrderDate    DATETIME2 NOT NULL DEFAULT SYSDATETIME(),  
92     CustomerId   INT NOT NULL,  
93     CONSTRAINT FK_Orders_Customers FOREIGN KEY (CustomerId)  
94         REFERENCES Customers (CustomerId)  
95 );  
96  
97 CREATE TABLE OrderItems (  
98     OrderItemId  INT IDENTITY(1,1) PRIMARY KEY,  
99     Quantity     INT NOT NULL,  
100    UnitPrice    DECIMAL(10, 2) NOT NULL,  
101    ProductId    INT NOT NULL,  
102    OrderId      INT NOT NULL,  
103    CONSTRAINT FK_OrderItems_Products FOREIGN KEY (ProductId)  
104        REFERENCES Products (ProductId),  
105    CONSTRAINT FK_OrderItems_Orders FOREIGN KEY (OrderId)  
106        REFERENCES Orders (OrderId)  
107 );  
108  
109 CREATE TABLE Reviews (  
110     ReviewId     INT IDENTITY(1,1) PRIMARY KEY,  
111     Rating       INT NOT NULL,  
112     Date         DATETIME2 NOT NULL DEFAULT SYSDATETIME(),  
113     Comment      NVARCHAR(1000) NULL,  
114     ProductId    INT NOT NULL,  
115     CustomerId   INT NOT NULL,  
116     CONSTRAINT FK_Reviews_Products FOREIGN KEY (ProductId)  
117         REFERENCES Products (ProductId),  
118     CONSTRAINT FK_Reviews_Customers FOREIGN KEY (CustomerId)  
119         REFERENCES Customers (CustomerId),  
120     CONSTRAINT CK_Reviews_Rating CHECK (Rating BETWEEN 1 AND 5)  
121 );  
122  
123 CREATE TABLE Payments (  
124     PaymentId    INT IDENTITY(1,1) PRIMARY KEY,  
125     PaymentDate  DATETIME2 NOT NULL,  
126     Amount       DECIMAL(10, 2) NOT NULL,  
127     Status       NVARCHAR(50) NOT NULL,  
128     Method       NVARCHAR(50) NOT NULL  
129 );
```

```
131 CREATE TABLE Orders_Payments (  
132     OrderId      INT NOT NULL,  
133     PaymentId    INT NOT NULL,  
134     CONSTRAINT PK_Orders_Payments PRIMARY KEY (OrderId, PaymentId),  
135     CONSTRAINT FK_Orders_Payments_Orders FOREIGN KEY (OrderId)  
136         REFERENCES Orders (OrderId),  
137     CONSTRAINT FK_Orders_Payments_Payments FOREIGN KEY (PaymentId)  
138         REFERENCES Payments (PaymentId)  
139 );  
140  
141 CREATE TABLE Shipments (  
142     ShipmentId    INT IDENTITY(1,1) PRIMARY KEY,  
143     ShipmentDate   DATETIME2 NOT NULL DEFAULT SYSDATETIME(),  
144     Status         NVARCHAR(50) NOT NULL,  
145     DeliveryDate   DATETIME2 NULL,  
146     CarrierName    NVARCHAR(100) NOT NULL,  
147     TrackingNumber NVARCHAR(100) NULL,  
148     OrderId        INT NOT NULL,  
149     CONSTRAINT FK_Shipments_Orders FOREIGN KEY (OrderId)  
150         REFERENCES Orders (OrderId)  
151 );  
152 GO  
153  
154 -- =====  
155 -- Sample data  
156 -- =====  
157  
158 INSERT INTO Categories (Name, Description, MainCategory) VALUES  
159     ('Electronics', 'Phones, laptops and other devices', NULL),  
160     ('Accessories', 'Cables, stands and other add-ons', NULL);  
161  
162 INSERT INTO Products (Name, UnitPrice, StockQuantity, Description, CategoryId) VALUES  
163     ('Dell Inspiron 15', 18500.00, 25, '15.6" laptop, 16GB RAM, 512GB SSD', 1),  
164     ('Samsung Galaxy A54', 9500.00, 40, '6.4" AMOLED, 128GB', 1),  
165     ('USB-C Cable', 85.00, 200, 'Braided 1m charging cable', 2),  
166     ('Phone Stand', 60.00, 150, 'Adjustable aluminium stand', 2),  
167     ('Laptop Sleeve 15"', 320.00, 45, 'Water resistant sleeve', 2);  
168  
169 -- Two movements before 2023 and two after, for the ArchivedStock operation.  
170 INSERT INTO StockTransactions (TranDate, QuantityChange, Type, Reference, ProductId) VALUES  
171     ('2021-05-10', 50, 'in', 1001, 1),  
172     ('2022-11-01', -10, 'out', 1002, 2),  
173     ('2023-04-20', 100, 'in', 1003, 3),  
174     ('2024-01-12', -5, 'out', 1004, 4);
```

```
176 INSERT INTO Customers (FullName, PhoneNumber, Email, ShippingAddress, RegistrationDate) VALUES
177     ('Ahmed Hassan', '+20 100 555 0001', 'ahmed.hassan@example.com', '12 Nasr Rd, Cairo', '2022-01-15'),
178     ('Mona Adel', '+20 101 555 0002', 'mona.adel@example.com', '5 Corniche, Alexandria', '2023-06-02');
179
180 -- Totals on both sides of 5000, for the Premium / Standard update.
181 INSERT INTO Orders (Status, TotalAmount, OrderDate, CustomerId) VALUES
182     ('Pending', 18500.00, '2024-03-01', 1),
183     ('Pending', 145.00, '2024-05-11', 1),
184     ('Pending', 9500.00, '2025-01-20', 2),
185     ('Pending', 320.00, '2025-02-14', 2);
186 GO
```

```
1  -- =====
2  -- Part 01 - Hotel Reservation Database
3  -- Builds the database from the provided relational schema, then adds a few
4  -- sample rows so the Part 02 operations have data to work on.
5  -- Typo fixes from the original diagram: ManageId -> ManagerId, SttafId -> StaffId, DaliyRate -> DailyRate.
6  -- Dialect: T-SQL (SQL Server)
7  -- =====
8
9  USE master;
10 GO
11
12 IF DB_ID('HotelReservation') IS NOT NULL
13 BEGIN
14     ALTER DATABASE HotelReservation SET SINGLE_USER WITH ROLLBACK IMMEDIATE;
15     DROP DATABASE HotelReservation;
16 END
17 GO
18
19 CREATE DATABASE HotelReservation;
20 GO
21
22 USE HotelReservation;
23 GO
24
25 -- ManagerId's FK to Staff is added after Staff exists below (circular reference: a hotel's
26 -- manager is a staff member, and every staff member works at one hotel).
27 CREATE TABLE Hotels (
28     HotelId      INT IDENTITY(1,1) PRIMARY KEY,
29     Name         NVARCHAR(150) NOT NULL,
30     Address      NVARCHAR(250) NOT NULL,
31     City         NVARCHAR(100) NOT NULL,
32     StarRating   INT NOT NULL,
33     ContactNumber NVARCHAR(20) NULL,
34     ManagerId    INT NULL UNIQUE,
35     CONSTRAINT CK_Hotels_StarRating CHECK (StarRating BETWEEN 1 AND 5)
36 );
37
38 CREATE TABLE Staff (
39     StaffId      INT IDENTITY(1,1) PRIMARY KEY,
40     FullName     NVARCHAR(150) NOT NULL,
41     Position     NVARCHAR(50) NOT NULL,
42     Salary       DECIMAL(10, 2) NOT NULL,
43     HotelId      INT NOT NULL,
44     CONSTRAINT FK_Staff_Hotels FOREIGN KEY (HotelId)
45         REFERENCES Hotels (HotelId)
46 );
```

```
48 ALTER TABLE Hotels
49     ADD CONSTRAINT FK_Hotels_Manager FOREIGN KEY (ManagerId)
50     REFERENCES Staff (StaffId);
51
52 CREATE TABLE Services (
53     ServiceId      INT IDENTITY(1,1) PRIMARY KEY,
54     ServiceName    NVARCHAR(100) NOT NULL,
55     Charge         DECIMAL(10, 2) NOT NULL,
56     RequestDate    DATETIME2 NOT NULL DEFAULT SYSDATETIME(),
57     StaffId        INT NOT NULL,
58     CONSTRAINT FK_Services_Staff FOREIGN KEY (StaffId)
59     REFERENCES Staff (StaffId)
60 );
61
62 CREATE TABLE Rooms (
63     RoomNumber     NVARCHAR(10) PRIMARY KEY,
64     RoomType       NVARCHAR(50) NOT NULL,
65     Capacity       INT NOT NULL,
66     DailyRate      DECIMAL(10, 2) NOT NULL,
67     Availability   NVARCHAR(50) NOT NULL,
68     HotelId        INT NOT NULL,
69     CONSTRAINT FK_Rooms_Hotels FOREIGN KEY (HotelId)
70     REFERENCES Hotels (HotelId)
71 );
72
73 CREATE TABLE Amenities (
74     RoomNumber     NVARCHAR(10) NOT NULL,
75     Amenity        NVARCHAR(50) NOT NULL,
76     CONSTRAINT PK_Amenities PRIMARY KEY (RoomNumber, Amenity),
77     CONSTRAINT FK_Amenities_Rooms FOREIGN KEY (RoomNumber)
78     REFERENCES Rooms (RoomNumber)
79 );
80
81 CREATE TABLE Reservations (
82     ReservationId  INT IDENTITY(1,1) PRIMARY KEY,
83     BookingDate    DATETIME2 NOT NULL DEFAULT SYSDATETIME(),
84     CheckInDate    DATE NOT NULL,
85     CheckOutDate   DATE NOT NULL,
86     ReservationStatus NVARCHAR(50) NOT NULL,
87     TotalPrice     DECIMAL(10, 2) NOT NULL,
88     NumberOfAdults INT NOT NULL,
89     NumberOfChildren INT NOT NULL DEFAULT 0,
90     CONSTRAINT CK_Reservations_Dates CHECK (CheckOutDate > CheckInDate)
91 );
```

```
93 CREATE TABLE Reservations_Rooms (  
94     ReservationId INT NOT NULL,  
95     RoomNumber NVARCHAR(10) NOT NULL,  
96     CONSTRAINT PK_Reservations_Rooms PRIMARY KEY (ReservationId, RoomNumber),  
97     CONSTRAINT FK_Reservations_Rooms_Reservations FOREIGN KEY (ReservationId)  
98         REFERENCES Reservations (ReservationId),  
99     CONSTRAINT FK_Reservations_Rooms_Rooms FOREIGN KEY (RoomNumber)  
100         REFERENCES Rooms (RoomNumber)  
101 );  
102  
103 CREATE TABLE ReservationService (  
104     ServiceId INT NOT NULL,  
105     ReservationId INT NOT NULL,  
106     CONSTRAINT PK_ReservationService PRIMARY KEY (ServiceId, ReservationId),  
107     CONSTRAINT FK_ReservationService_Services FOREIGN KEY (ServiceId)  
108         REFERENCES Services (ServiceId),  
109     CONSTRAINT FK_ReservationService_Reservations FOREIGN KEY (ReservationId)  
110         REFERENCES Reservations (ReservationId)  
111 );  
112  
113 CREATE TABLE Guests (  
114     GuestId INT IDENTITY(1,1) PRIMARY KEY,  
115     FullName NVARCHAR(150) NOT NULL,  
116     Nationality NVARCHAR(100) NULL,  
117     PassportNumber NVARCHAR(50) NOT NULL UNIQUE,  
118     DateOfBirth DATE NOT NULL  
119 );  
120  
121 CREATE TABLE Guest_Contact_Details (  
122     GuestId INT NOT NULL,  
123     Detail NVARCHAR(150) NOT NULL,  
124     CONSTRAINT PK_Guest_Contact_Details PRIMARY KEY (GuestId, Detail),  
125     CONSTRAINT FK_Guest_Contact_Details_Guests FOREIGN KEY (GuestId)  
126         REFERENCES Guests (GuestId)  
127 );  
128  
129 CREATE TABLE Reservations_Guest (  
130     ReservationId INT NOT NULL,  
131     GuestId INT NOT NULL,  
132     CONSTRAINT PK_Reservations_Guest PRIMARY KEY (ReservationId, GuestId),  
133     CONSTRAINT FK_Reservations_Guest_Reservations FOREIGN KEY (ReservationId)  
134         REFERENCES Reservations (ReservationId),  
135     CONSTRAINT FK_Reservations_Guest_Guests FOREIGN KEY (GuestId)  
136         REFERENCES Guests (GuestId)  
137 );
```

```

139 CREATE TABLE Payments (
140     PaymentId      INT IDENTITY(1,1) PRIMARY KEY,
141     Method          NVARCHAR(50) NOT NULL,
142     Date            DATETIME2 NOT NULL,
143     Amount          DECIMAL(10, 2) NOT NULL,
144     ConfirmationNumber NVARCHAR(50) NULL
145 );
146
147 CREATE TABLE Reservations_Payment (
148     ReservationId   INT NOT NULL,
149     PaymentId       INT NOT NULL,
150     CONSTRAINT PK_Reservations_Payment PRIMARY KEY (ReservationId, PaymentId),
151     CONSTRAINT FK_Reservations_Payment_Reservations FOREIGN KEY (ReservationId)
152         REFERENCES Reservations (ReservationId),
153     CONSTRAINT FK_Reservations_Payment_Payments FOREIGN KEY (PaymentId)
154         REFERENCES Payments (PaymentId)
155 );
156 GO
157
158 -- =====
159 -- Sample data
160 -- =====
161
162 INSERT INTO Hotels (Name, Address, City, StarRating, ContactNumber) VALUES
163     ('Nile Grand Hotel', '1 Corniche El Nil', 'Cairo', 5, '+20 2 2555 1000');
164
165 INSERT INTO Rooms (RoomNumber, RoomType, Capacity, DailyRate, Availability, HotelId) VALUES
166     ('101', 'Standard', 2, 1200.00, 'Available', 1),
167     ('102', 'Deluxe', 3, 1800.00, 'Available', 1),
168     ('201', 'Suite', 4, 3500.00, 'Occupied', 1),
169     ('202', 'Suite', 4, 3800.00, 'Available', 1);
170
171 -- Dates are relative to today, so the status update always finds a finished,
172 -- an ongoing and a future stay.
173 INSERT INTO Reservations (CheckInDate, CheckOutDate, ReservationStatus, TotalPrice, NumberOfAdults, NumberOfChildren) VALUES
174     (CAST(DATEADD(DAY, -30, GETDATE()) AS DATE), CAST(DATEADD(DAY, -25, GETDATE()) AS DATE), 'Confirmed', 6000.00, 2, 0),
175     (CAST(DATEADD(DAY, -2, GETDATE()) AS DATE), CAST(DATEADD(DAY, 3, GETDATE()) AS DATE), 'Confirmed', 17500.00, 2, 1),
176     (CAST(DATEADD(DAY, 20, GETDATE()) AS DATE), CAST(DATEADD(DAY, 24, GETDATE()) AS DATE), 'Confirmed', 7200.00, 3, 0);
177 GO

```

```
1  -- =====
2  -- Part 02 - Online Retail System: Insert & Update Operations
3  -- Run after Part_01/OnlineRetailStore.sql.
4  -- =====
5
6  USE OnlineRetailStore;
7  GO
8
9  -- =====
10 -- 1. INSERT OPERATIONS
11 -- =====
12
13 -- Insert a new Customer
14 INSERT INTO Customers (FullName, PhoneNumber, Email, ShippingAddress, RegistrationDate)
15 VALUES ('Nada Farouk', '+20 106 555 0006', 'nada.farouk@example.com', '45 El Orouba St, Heliopolis', GETDATE());
16
17 -- Insert 3 new Suppliers
18 INSERT INTO Suppliers (Name, Country, Email, Address, ContactNumber) VALUES
19     ('Nile Tech Distribution', 'Egypt', 'sales@niletech.com', '14 El Tahrir St, Cairo', '+20 100 111 2222'),
20     ('Cairo Home Supplies', 'Egypt', 'info@cairohome.com', '8 Gesr El Suez, Cairo', '+20 101 333 4444'),
21     ('Global Gadgets Ltd', 'China', 'contact@globalgadgets.cn', '221 Huaqiang Rd, Shenzhen', '+86 755 8888 9999');
22
23 -- Insert 2 Categories
24 INSERT INTO Categories (Name, Description) VALUES
25     ('Gaming', 'Consoles, controllers and gaming gear'),
26     ('Wearables', 'Smart watches and fitness bands');
27
28 -- Insert a Product with only Name and UnitPrice
29 -- StockQuantity and AddedDate take their defaults; Description and CategoryId stay NULL.
30 INSERT INTO Products (Name, UnitPrice)
31 VALUES ('Wireless Charger 15W', 450.00);
32
33 -- Create ArchivedStock and fill it with every stock transaction before 2023
34 CREATE TABLE ArchivedStock (
35     TranId          INT          PRIMARY KEY,
36     ProductId       INT          NOT NULL,
37     QuantityChange  INT          NOT NULL,
38     TranDate        DATETIME2    NOT NULL
39 );
40
41 INSERT INTO ArchivedStock (TranId, ProductId, QuantityChange, TranDate)
42 SELECT TranId, ProductId, QuantityChange, TranDate
43 FROM StockTransactions
44 WHERE TranDate < '2023-01-01';
```

```
46 SELECT * FROM ArchivedStock;
47 GO
48
49 -- =====
50 -- 2. UPDATE OPERATIONS
51 -- =====
52
53 -- Increase UnitPrice by 10% for every product under 100 EGP
54 UPDATE Products
55 SET UnitPrice = UnitPrice * 1.10
56 WHERE UnitPrice < 100;
57
58 -- Set each order's status from its total amount
59 UPDATE Orders
60 SET Status = CASE
61     WHEN TotalAmount > 5000 THEN 'Premium'
62     ELSE 'Standard'
63 END;
64
65 SELECT ProductId, Name, UnitPrice FROM Products;
66 SELECT OrderId, TotalAmount, Status FROM Orders;
67 GO
```

```
1  -- =====
2  -- Part 02 - Hotel System: Insert & Update Operations
3  -- Run after Part_01/HotelReservation.sql.
4  -- =====
5
6  USE HotelReservation;
7  GO
8
9  -- =====
10 -- 1. INSERT OPERATIONS
11 -- =====
12
13 -- Insert a Guest
14 INSERT INTO Guests (FullName, Nationality, PassportNumber, DateOfBirth)
15 VALUES ('John Smith', 'American', 'US1234567', '1985-04-12');
16
17 -- Insert multiple Guests in one statement
18 INSERT INTO Guests (FullName, Nationality, PassportNumber, DateOfBirth) VALUES
19     ('Maria Garcia', 'Spanish', 'ES9988776', '1990-11-03'),
20     ('Ali Mansour', 'Egyptian', 'EG5544332', '1978-07-22'),
21     ('Chen Wei', 'Chinese', 'CN7766554', '1995-01-30');
22
23 SELECT GuestId, FullName, Nationality, PassportNumber, DateOfBirth FROM Guests;
24 GO
25
26 -- =====
27 -- 2. UPDATE OPERATIONS
28 -- =====
29
30 -- Increase DailyRate by 15% for all suites
31 UPDATE Rooms
32 SET DailyRate = DailyRate * 1.15
33 WHERE RoomType = 'Suite';
34
35 -- Set each reservation's status from its dates
36 UPDATE Reservations
37 SET ReservationStatus = CASE
38     WHEN CheckOutDate < GETDATE() THEN 'Completed'
39     WHEN CheckInDate > GETDATE() THEN 'Upcoming'
40     ELSE 'Active'
41 END;
42
43 SELECT RoomNumber, RoomType, DailyRate FROM Rooms;
44 SELECT ReservationId, CheckInDate, CheckOutDate, ReservationStatus FROM Reservations;
45 GO
```